ADSORPTION, DESORPTION, AND SURFACE REACTIONS

H.P. BONZEL

Institut für Grenzflächenforschung und Vakuumphysik
Kernforschungsanlage Jülich GmbH, Postfach 1913, D-5170 Jülich
Federal Republic of Germany

1. INTRODUCTION

The interaction of atoms and molecules with solid surfaces can be broken up into several fundamental processes, such as adsorption, surface diffusion, surface reaction, and desorption. The details of these fundamental processes have been investigated in numerous studies during recent years. Many of these investigations have been carried out on well defined single crystal surfaces utilizing various surface analytical tools and modern ultra-high vacuum technology. The use of single crystal surfaces is of particular importance in these studies in order to relate the physical characteristics of the fundamental processes mentioned above to specific adsorption sites.

The experimental investigations can be classified into two different groups. The first group consists of dynamic experiments concerned with the rates of adsorption and desorption of gases and molecules. The species involved are usually analyzed in the gas phase by mass spectrometry. The other type of experiment concerns the stationary adsorbed phase on the surface which is analyzed by various electron spectroscopies and additional scattering experiments involving atoms and ions. The goal of the latter experiments is to determine the physical properties of the adsorbate, such as the adsorption energy, adsorption site, the substrate molecule bond including bond length and molecule orientation, the ordering tendency of the adsorbate, its vibrational properties, and its propensity of dissociation. Of further interest are adsorbate-induced changes in substrate atom-atom bonding which may change the degree of relaxation between surface layers, eventually leading to surface reconstruction. In this brief review we will emphasize the dynamical aspects of the experiments and only give a few examples of investigations of the stationary adsorbed phase.

2. ADSORPTION

2.1. Kinetics of adsorption

The rate of adsorption of molecules on a solid surface depends on several parameters and is given by the following equation [1,2]:

$$r_a = \frac{dn}{dt} = J\, s(n,T) = s_o\, f(n)\, F_1 \exp\left(-\frac{E}{kT}\right) \qquad (1)$$

In this equation, n is the number of adsorbed molecules/cm^2, J the impingement flux (molecules/cm^{2}s), $s(n,T)$ the sticking probability, s_o the initial sticking coefficient, $f(n)$ a function describing the coverage dependence of sticking, F_1 an entropy term, E the activation energy of adsorption, k the Boltzmann constant and T the absolute temperature of the gas molecules. At this point we can distinguish several different kinds of adsorption processes. Depending on the magnitude of E, one speaks of activated or non-activated adsorption (E=0). It is also important to know

L. D. Laude et al. (eds.), *Interfaces Under Laser Irradiation*
© Springer Science+Business Media Dordrecht 1987

whether the adsorption will be molecular or dissociative. In the simplest case of molecular adsorption the site function f(n) will be $(1-n/n_0)$ where n_0 is the number of sites per cm^2. In that case, one speaks of Langmuir adsorption. The energetics of adsorption are illustrated in Fig.1 in simple terms by a one-dimensional adsorption potential [3]. The left-hand figure

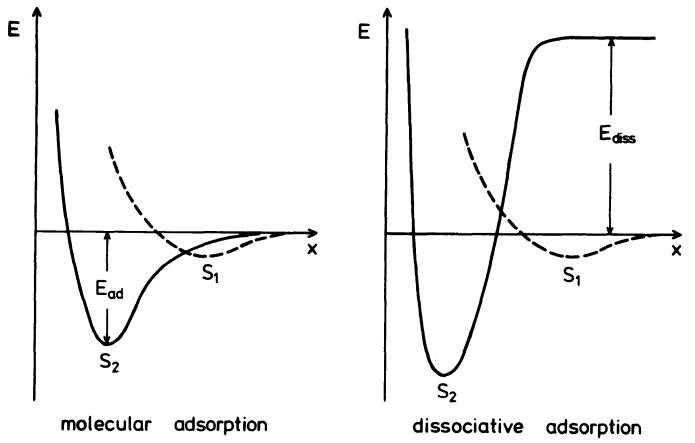


FIGURE 1: One-dimensional adsorption potential for molecular (left) and dissociative (right) adsorption. The activation barrier for adsorption results from the crossing point of the potentials for S_1 and S_2 above the x-axis (E >0).

shows the situation for molecular adsorption featuring a potential for a weakly held state, S_1, and a second potential for a more strongly held state, S_2 whose energy of adsorption is E_{ad}. The weakly held state S_1 is due to dispersion forces (physisorbed state) and is believed to be a precursor for the more tightly bound chemisorbed state S_2. There is no activation barrier for the transition from S_1 to S_2. The right-hand side of Fig. 1 shows the situation for a dissociative adsorption with a finite activation barrier between the first state S_1 and the tightly held dissociated state S_2. At large distances from the surface there is an energetic separation between the two potentials, the energy of dissociation of the molecule in the gas phase. Activated adsorption is generally only observed in context with dissociative adsorption. The first adsorbed state S_1 may be a weakly held precursor state to dissociation or a tightly bound, chemisorbed molecular state.

The measured quantity characterizing the rate of adsorption is the temperature and coverage dependent sticking coefficient defined by equation (1):

$$s(n,T) = \frac{1}{J}\frac{dn}{dt} \qquad (2)$$

As the impingement flux J is known from gas kinetic theory

$$J = \frac{P}{\sqrt{2\pi m\, kT}} \qquad (3)$$

the rate of adsorption dn/dt must be determined experimentally. There are several different techniques to measure dn/dt. In the first, the concentration of the adsorbate n is measured as a function of t by some kind of surface analytical technique. This results in the so-called uptake curve n(t). The sticking coefficient is then obtained by taking the derivative of this curve and dividing by the impingement flux. In the second approach the coverage is measured by a quantitative desorption experiment. The temperature of the crystal is raised, and the partial pressure P(t) of the desorbing gas is monitored as a function of temperature by a mass spectrometer or an ion gauge. As the experiment is performed in a continuously pumped system, the partial pressure is proportional to the desorption rate, and the area under the desorption trace is a measure of the coverage of the adsorbed species prior to the flash experiment (compare section 4.1.). The third approach is based on measuring the intensity of a molecular beam scattered at the surface [4]. At the beginning of the experiment the surface is clean and therefore adsorbs molecules with a relatively large rate. Thus, the intensity of the scattered beam is low. With increasing coverage the rate of adsorption decreases leading to an increase in the scattered intensity of the molecular beam. The sticking coefficient is therefore simply the difference between the incident intensity and the scattered intensity of the molecular beam divided by the incident intensity:

$$s(t) = \frac{I_0 - I(t)}{I_0} = \frac{I_\infty - I(t)}{I_\infty} \qquad (4)$$

The incident intensity I_0 is usually replaced by the intensity of the scattered beam for a saturated surface, I_∞. It is also important that the scattered intensities are integrated over all angles. The coverage is calculated by integrating over the sticking coefficient.

In the following we discuss two examples of measurements of the rate of adsorption, one for reversible molecular adsorption and the other for dissociative adsorption. The first example concerns the adsorption of CO on Pt(111). In this case the uptake of CO on Pt was obtained by measuring the adsorbed amount in a flash desorption experiment [5] . Figure 2 shows desorption traces of CO from Pt(111) for various initial coverages. The adsorbed amount is determined quantitatively when the mass spectrometer is calibrated and the pumping speed of the system is known. The uptake curve of CO determined in this way is shown in Fig. 3. Here the coverage in molecules/cm^2 is plotted versus the total CO exposure in molecules/cm^2. The sticking coefficient is simply the derivative of this function which is shown in Fig. 4 for the adsorption temperature of 300 K. Apart from the initial rise at very low coverage which is not understood at this time, the sticking coefficient is roughly constant at 0.8 up to a coverage of about 0.2 and then decreases with increasing coverage until it reaches a very low value at a coverage of half a monolayer. The carbon monoxide molecules form an ordered layer at this coverage such that all ordinary adsorption sites are filled [6]. Fig. 4 shows for comparison also a measurement of the sticking coefficient by the molecular beam technique [7]. The substrate temperature is 310 K. Although the absolute values of the two measurements are quite comparable, the functional behaviour is somewhat different. The data of the molecular beam experiment obey quite well the Kisliuk model which we will describe very briefly.

The Kisliuk model [8,9] deals with a particular form of the site function f(n). It is based on a molecular adsorption potential as shown in Fig. 1. Two different precursor sites are distinguished, one located over

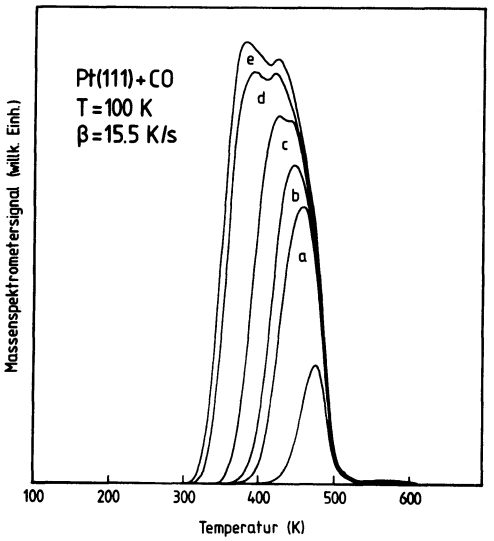


FIGURE 2: Desorption spectra of CO from Pt(111) for increasing initial coverages. CO was adsorbed at 100 K [5].

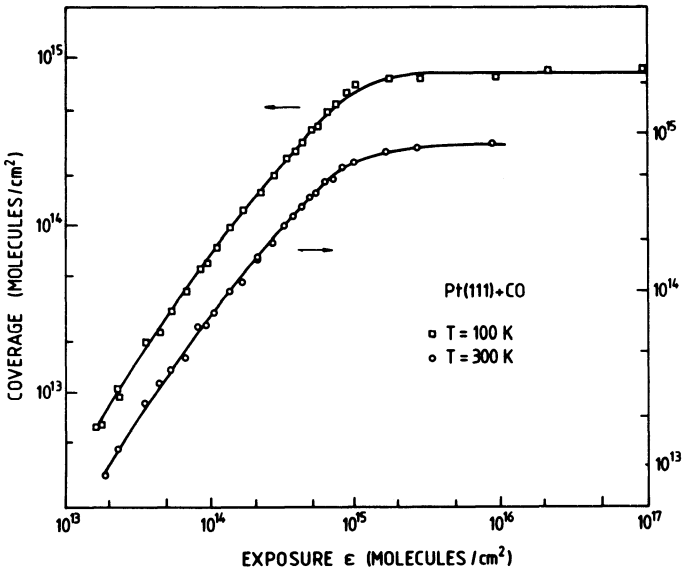


FIGURE 3: Plot of CO coverage versus CO exposure at 100 and 300 K [5].

an empty chemisorption site and the other over a filled chemisorption site. Free exchange between the two precursor sites via surface diffusion is allowed. Chemisorption can occur only from the precursor located over the

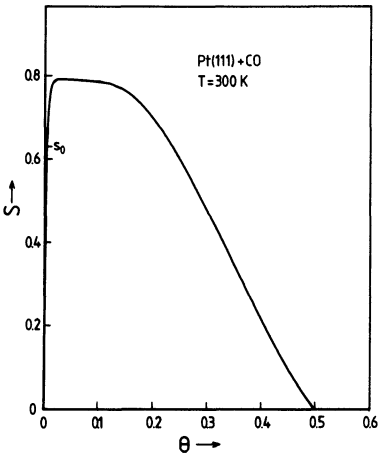
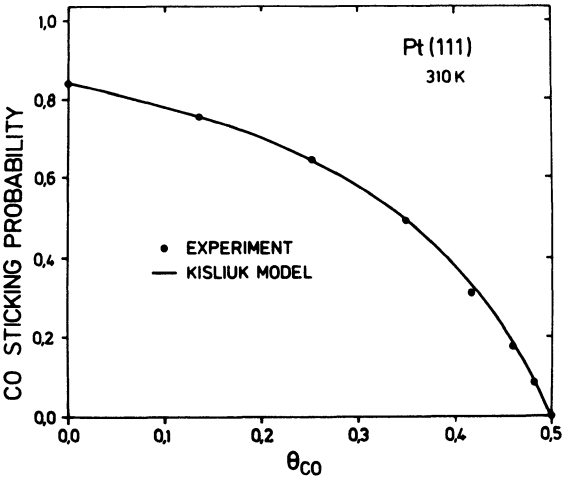


FIGURE 4: (a) Coverage dependence of sticking coefficient of CO on Pt(111) at 300 K [5]. (b) Same at 310 K but obtained by molecular beam technique [7].



empty site. This model has had some success in explaining experimental data but this is perhaps not surprising because of a relatively large number of fitting parameters.

In the second example we present some data for the dissociative chemisorption of H_2 on several Ni surfaces. It was found that the adsorption of H_2 on Ni was structure-sensitve in the sense that the close-packed Ni(111) surface showed much less reactivity than the more open Ni(110) surface. The sticking coefficient for hydrogen on these two surfaces at a sample temperature of 140 K is presented in Fig. 5 [10]. The data illustrate the strikingly different behavior of the sticking coefficient with coverage for these two surfaces. Whereas the initial sticking coefficient on Ni(111) is only 0.05, it is 0.96 for the Ni(110) surface. Furthermore, the sticking coefficient on Ni(111) decreases rapidly with increasing coverage towards very low values while it stays constant on Ni(110) up to coverages of about 0.5 and then decreases up to a coverage of 1.0. The data for both surfaces are well described by a modified Kisliuk model [10]. Such low sticking coefficients as observed for H_2 on Ni(111) are indicative of activated

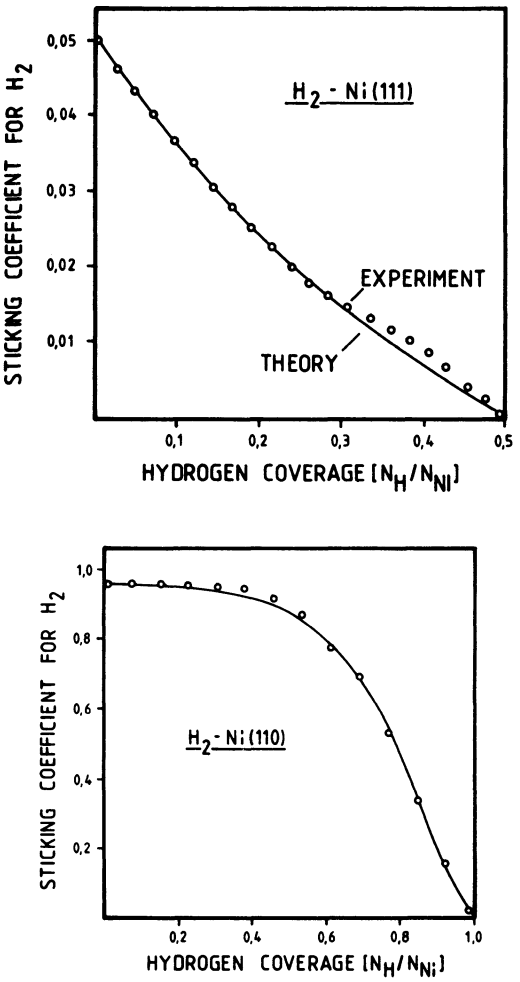


FIGURE 5: Coverage dependence of sticking coefficient of H_2 on Ni(111) and Ni(110) at 140 K. The solid lines are fitted to the data on the basis of an adsorption model [10].

adsorption and call for an increase in the gas temperature such that the activation energy can be surmounted by the impinging molecules. Such an experiment has actually been carried out by utilizing molecular beam techniques [11] or a heated Knudsen cell [12]. Both experiments report a considerable increase in the initial sticking coefficient of H_2 on Ni(111) with increasing gas temperature, as shown in Fig. 6 although the functional dependence is not the same. The same effect cannot be achieved by raising the surface temperature of the Ni crystal [12].

Structure sensitivity in adsorption is observed quite frequently. The simplest kind is the sensitivity of the rate of adsorption to the crystallographic surface orientation, as we saw in the example of H_2 on Ni. Mostly structure sensitivity is only pronounced with dissociative adsorption.

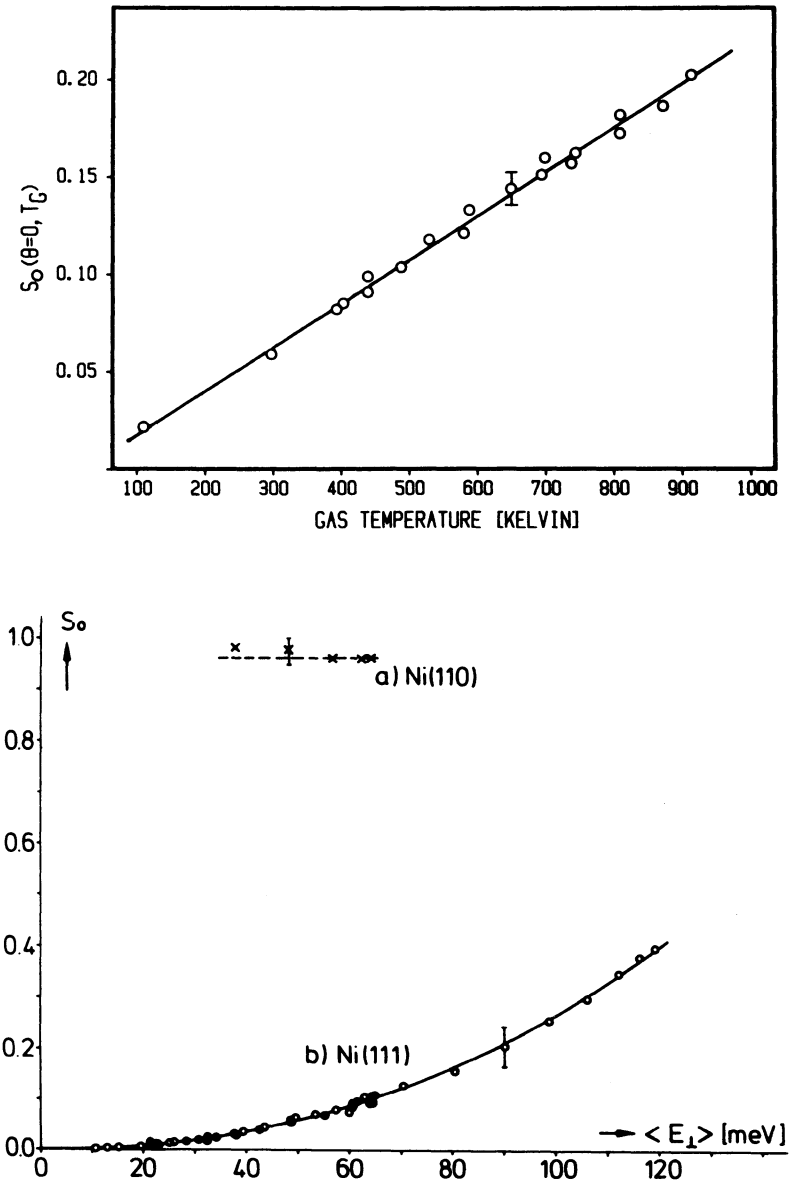


FIGURE 6: Variation of the initial sticking coefficient of H_2 on Ni(111) with gas temperature (= <E>/2k). (a) H_2 emitted from a heated Knudsen cell, Ni crystal at 190 K [12]. (b) with H_2 molecular beam [11].

Other well known examples are N_2 [13] and H_2 on W surfaces [14]. A more subtle case of structure sensitivity has to do with reconstructed surfaces. For example, the adsorption of O_2 on a non-reconstructed Pt(100) 1x1 sur-

face is considerably faster at room temperature than on the reconstructed Pt(100) 5x20 surface [15]. Also, the dissociative adsorption of NO on the same 1x1 surface is more rapid than on the (5x20) modification [16]. In general, surface roughness has a tendency to increase the rate of adsorption [17]. This effect has been studied by using surfaces with a distribution of monatomic steps. It was found that the initial rate of adsorption increased with the density of steps, for example, for O_2 on Si [18], H_2 on Pt(111) [19], and O_2 on W(110) [20]. Finally it should be mentioned in this context that surface additives, such as foreign adsorbed atoms, can have a substantial influence on the rate of gas adsorption [21]. Electropositive additives (alkali metals) quite often increase the rate of adsorption while electronegative additives (S,P,C,O) have a tendency to decrease the rate of adsorption. These effects, which are important in heterogeneous catalysis, have been studied in great detail for the adsorption of CO, O_2 and other simple gases.

2.2. Energy of adsorption

The energy of adsorption is an important quantity which characterizes the interaction between the adsorbed molecule and surface atoms of the substrate. There are several possible ways of obtaining the energy of adsorption. One way is by means of calorimetry, but this technique does not work successfully with small-area single crystal surfaces. A second popular technique is by means of isobars or isotherms [22]. In this approach, the surface coverage of adsorbed molecules is measured as a function of the sample temperature at constant pressure in the gas phase (isobar). An alternate way is to measure the coverage as a function of pressure at constant sample temperature (isotherm). If several of these curves are obtained for different pressures (different temperatures), the data can be cross-plotted as log P versus 1/T for constant coverage of the adsorbate (isostere). This plot should result in a number of straight lines whose slopes are the isosteric heat of adsorption. The evaluation is according to the Clausius-Clapeyron equation, as follows:

$$\left.\frac{d\ln P}{d(1/T)}\right|_{\Theta} = -\frac{E_{ad}(n)}{k} \qquad (5)$$

This type of measurement works very well for single crystal surfaces as long as the coverage of the adsorbate can be measured in-situ by a suitable surface analytical technique. Quite often the coverage is monitored by a change in the work function. A typical result for E_{ad} (n) is shown in Fig. 7 for CO on Pt(111) [6]. Here, a general decrease of the adsorption energy with increasing coverage is observed, a behaviour which indicates repulsive interaction between adsorbed CO molecules.

A third way of obtaining the energy of adsorption is by measuring the lifetime of the adsorbed species with a modulated molecular beam. Here, the lifetime is related to the reciprocal rate of desorption which actually yields the energy of desorption. However, for non-activated adsorption, the energy of adsorption is equal to the energy of desorption. This type of measurement has been carried out for CO on Pt(111) resulting in E_{ad} = 34.9 kcal/mol [7]. Note that the technique measures E_{ad} for very low coverages of CO and since the Pt(111) surface contained steps, the energy of adsorption is characteristic of CO adsorbed at step sites. Therefore E_{ad} determined here is higher than that in Fig. 7 for low CO coverage.

3. CHARACTERIZATION OF ADSORBED MOLECULES

As already outlined in the introduction, adsorbed molecules are investigated by a number of surface-spectroscopic and surface-scattering

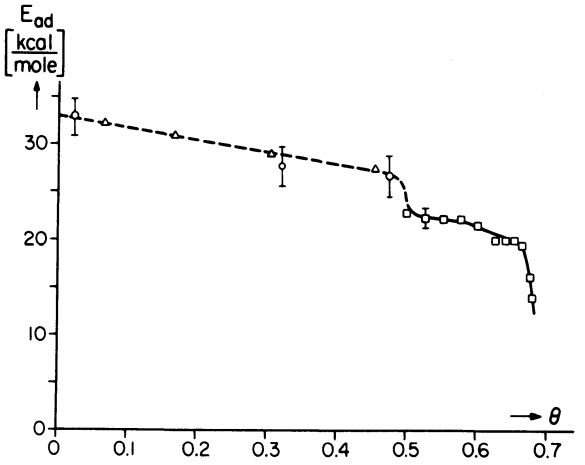


FIGURE 7: Adsorption energy of CO on Pt(111) as a function of CO coverage [6].

techniques. The general goal is a physical characterization of the adsorbed species as well as the adsorbate-substrate bonding. The physical properties are studied as a function of coverage and on different crystallographic surface orientations, such that the site specificity of adsorption can be elucidated.

Many adsorbed species have a tendency to form ordered layers on the surface. Depending on the kind of interaction between the adsorbed species, this ordering can occur at low or at high coverage. The most common tool to study ordering is low energy electron diffraction (LEED). For example, LEED studies of CO adsorption on Pt(111) have shown a whole series of ordered structures with increasing CO coverage [5].

When the energy of adsorption is very large, the presence of adsorbed species on the surface represents a major perturbation of the interatomic bonding near the surface. This alters bond distances as well as the separation between successive layers of the solid. In some cases the perturbation of the energetic equilibrium at the surface can be so dramatic that the structure of the surface changes (reconstruction). Several of these adsorbate-induced reconstructions of metal surfaces are known. Examples are O_2 and H_2 on Ni(110) [23,24] and Cu(110) [25] surfaces, and Na on Ag(110) [26] surfaces. Elaborate experimentation was necessary in each of these cases, in order to verify surface reconstruction and, in particular, to clarify the structure in each case.

Of great interest is the knowledge of the adsorption site in which the molecule resides on the surface. Several techniques are suitable for solving this problem. Particularly successful are vibrational spectroscopies where the measured vibrational frequencies of adsorbed molecules have been linked to the geometry the adsorption site. Symmetry selection rules and vibrational frequency shifts both serve to identify the adsorption site [27]. More about this problem will be said in detail in the lecture by Benedek (this volume).

A further question concerns the orientation of the molecule relative to the surface. Although vibrational spectra can also yield information on this question, other techniques such as the study of the near-edge structure of absorption spectra (NEXAFS) [28] or X-ray photoelectron diffraction (XPD) [29] are specifically suited to address this problem. As an example

we show in Fig. 8 electron yield spectra of CO on Pt(111) [30]. Here, the partial electron yield is plotted versus energy of the incident photons which are absorbed to produce secondary electrons. The upper part of the figure shows the carbon K edge structure whereas the lower part shows the

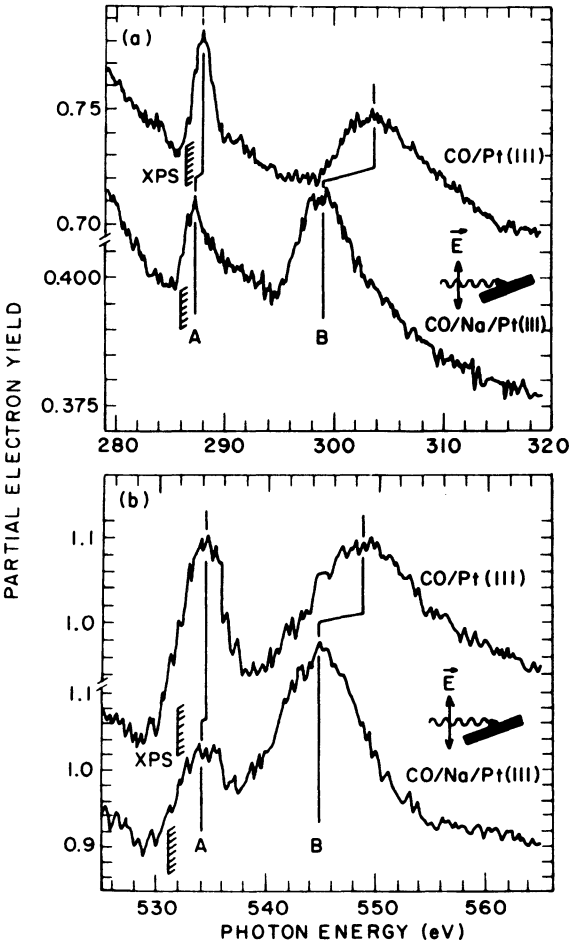


FIGURE 8: NEXAFS spectra for CO on clean and Na-covered Pt(111) recorded at 20° grazing X-ray incidence. Peak A is a π resonance and peak B a σ shape resonance. XPS denotes the absorption edge of the C1s(O1s) level. (a) Carbon K edge spectra. (b) Oxygen K edge spectra [30].

oxygen K edge structure. All spectra are characterized by a narrow π resonance and a broad σ resonance. The energetic separation between the π and σ resonances can be evaluated to yield the bond length of the adsorbed CO molecule. Furthermore, the polarization dependence of the π and σ resonance intensities tells the experimenter the orientation of the CO molecule relative to the surface. For example, the σ resonance is only observed when

the electric field vector of the incident photon radiation is close to parallel to the CO bond axis. The result of the experiment for CO on clean Pt(111) is that CO molecules are perpendicularly adsorbed on the Pt surface, with the carbon atom closest to the surface. The same experiment was also carried out for a Na-covered Pt(111) surface. Fig. 8 shows in this case that the energetic separation between π and σ resonance has decreased by about 4 eV. This decrease corresponds to an increase in the CO bond length of 0.12 Å, which shows that coadsorbed alkali metals can drastically alter the intramolecular bond of the CO molecule, and therefore also the adsorption bond of CO.

Another way of obtaining the orientation of a molecule relative to a surface is by X-ray photoelectron diffraction. Here the emission of photoelectrons from strongly bound core levels is measured as a function of polar angle for a certain azimuth. Fig. 9 shows again as an example CO on clean Pt(111) [31]. The polar angle dependence of the C1s and O1s photoelectron emission is plotted in the upper part of the figure and shows a peak near normal in the C1s intensity. This peak is enhanced by taking the ratio of carbon to oxygen emission shown in the lower part of Fig. 9. The

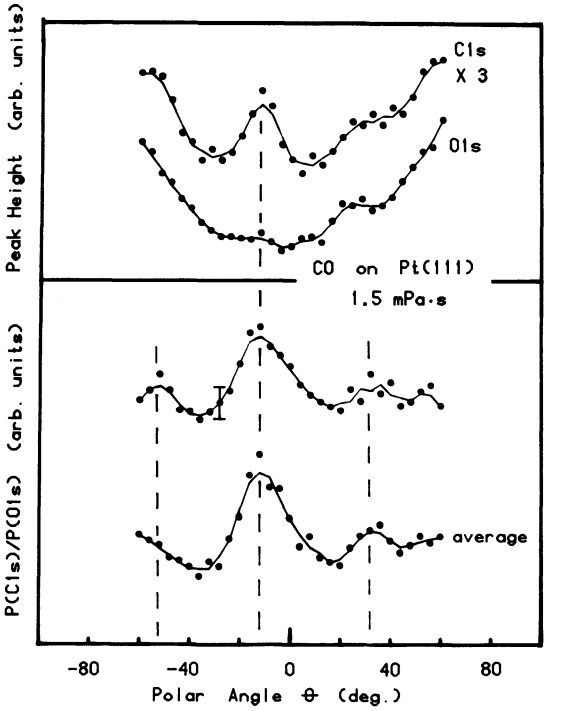


FIGURE 9: Polar angle dependence of C1s and O1s core level emission of CO adsorbed on Pt(111). Upper part shows intensities after background subtraction; lower part shows the ratio of peak heights; data labelled "average" correspond to the average of 3 independent measurements [31].

peak in the C1s emission occurs because the photoelectrons emitted from the carbon atom are strongly forward scattered by the oxygen atom. This peak then tells us two things: first, the CO molecule is adsorbed perpendicular to the surface, and secondly it is bonded with the carbon atom

down. By comparison, the C1s/O1s dependence for CO on a K-covered Pt(111) surface shows a very similar dependence such that the mode of CO bonding does not seem to be altered by the presence of K atoms [31].

The electronic characterization of adsorbed molecules is usually carried out by photoelectron spectroscopy (UPS, XPS). The objective is to measure the binding energy of electrons in the valence band region (molecular orbitals) as well as the core level region. These binding energies are then compared to those of gas phase molecules such that energy shifts can be identified. These energy shifts often indicate molecular orbitals which are involved in the adsorption bond. The same technique is also extremely valuable for distinguishing molecular and dissociative adsorption. In many cases, it is quite easy on the basis of identified molecular orbital peaks to say whether the adsorbed species is in its original molecular form or whether it has dissociated on adsorption. The same kind of identification can be carried out by means of binding energy shifts of core levels. Fig. 10 shows as an example UPS spectra of CO adsorbed on clean and K covered Fe(110) at room temperature and also at elevated temperatures [32]. The room temperature spectra are characterized by the familiar molecular

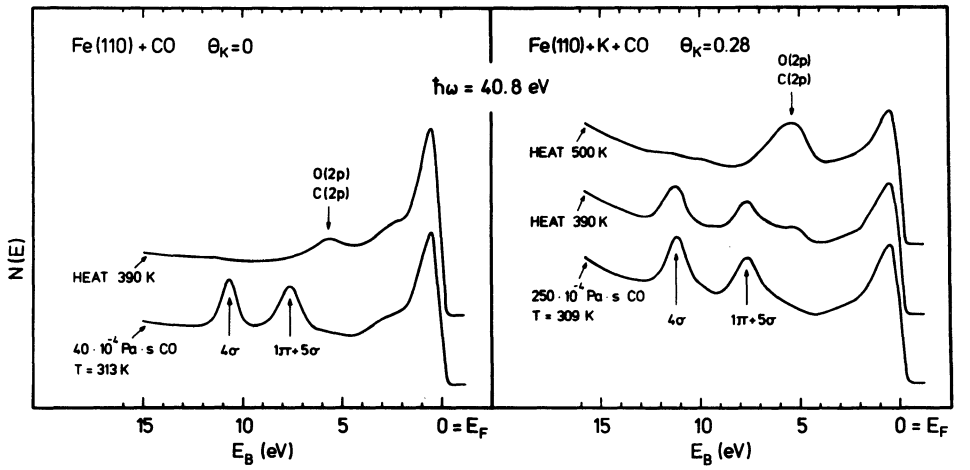


FIGURE 10: UPS spectra of CO adsorbed on clean and K-covered Fe(110): after adsorption at about 310 K and after heating to 390 K and 500 K, respectively [32].

orbital levels of CO (5σ, 1π, 4σ) which by their very presence tell us that CO is adsorbed as a molecule. On the other hand, spectra taken at the higher temperatures for the clean Fe(110) surface do not exhibit the molecular orbitals of CO but rather evidence for the atomic-like orbitals of oxygen and carbon, thus indicating the dissociation of CO. On the K-covered surface, on the other hand, heating to 390 K results in a mixture of molecular and dissociated CO as seen by the spectrum on the right hand side of the figure. From this we conclude that $E_{ad}(CO)$ is increased by the presence of K. Furthermore, a comparison of the clean and K-covered Fe(110) surface shows that CO dissociation is more pronounced on the K-covered surface. Thus, the total amount of dissociated CO is larger on that surface, an observation which can be readily explained by the higher energy of adsorption of CO on the K-covered Fe surface.

A final remark concerns the rate of surface diffusion of adsorbed

species. This process is particularly difficult to analyze because a spectroscopy with a very high lateral resolution is needed. One such technique is the field electron microscope with which the diffusion of adsorbed species across a metal surface has been studied [33]. Another very useful technique should be the newly developed scanning electron tunneling micro scope [34]. A completely different approach involves the scattering of helium beams from surfaces [35]. Here, the scattered beam intensity is a measure of the coverage of adsorbed molecules on flat parts of the surface. If molecules are adsorbed at low temperature, sufficiently low such that no surface diffusion takes place, the intensity of the scattered beam decreases. As the temperature is raised and the adsorbed species begin to diffuse to sites of higher adsorption energy (step sites, defect sites), the scattered helium beam intensity starts to increase again. Thus the increasing helium intensity is linked to the onset of surface diffusion of adsorbed species. Generally speaking, there is a great lack of information on surface diffusion data of adsorbed atoms and molecules on solid surfaces. Most recently, NMR has been applied to the measurement of surface diffusion of adsorbed molecules on metallic clusters [36].

4. DESORPTION

4.1. Rate of desorption

The desorption rate of atoms and molecules from surfaces is generally assumed to be described by the following equation [1,37]:

$$r_d = \frac{dn}{dt} = n^x \, k_{ox} \exp\left(-\frac{E_d(n)}{kT}\right) \tag{6}$$

In this equation x is the order of the desorption reaction, k_{ox} is the pre-exponential factor which includes an entropy term, and E_d is the energy of desorption which is generally coverage dependent. Strictly speaking, this equation cannot be applicable if the adsorbed species are residing in different adsorption sites with different adsorption energies. However, this difference is usually approximated by the coverage dependent adsorption energy. In a typical desorption experiment, the temperature of the substrate is raised in a linear fashion, such as

$$T(t) = T_0 + \beta t \tag{7}$$

where β is the heating rate. At a temperature where the desorption of the adsorbed species begins, the partial pressure of this particular species rises as shown in Fig. 2. If the experiment is carried out in a continuously pumped vacuum chamber, the pressure rise due to desorption will only be temporary because the amount of adsorbed gas is finite. Therefore the partial pressure of the desorbing gas as a function of temperature (time) will exhibit a maximum. As the partial pressure of the gas is directly proportional to the rate of desorption, the position of the maximum can be obtained from differentiating equation (6). The result is a condition given by the following equation [38]:

$$\frac{E_d}{kT_p^2} = x \, n^{x-1} \frac{k_{ox}}{\beta} \exp\left(-\frac{E_d}{kT_p}\right) \tag{8}$$

which relates the desorption energy to the temperature at the maximum, T_p. This equation is generally very useful for estimating E_d of the desorbing

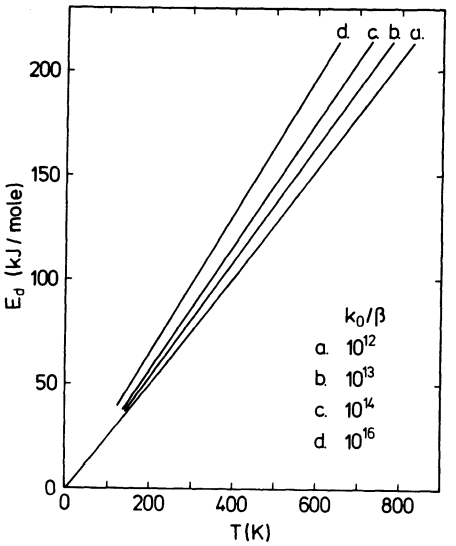


FIGURE 11: Plot of energy of desorption versus peak temperature of first order flash desorption traces, according to eq. (8) [38].

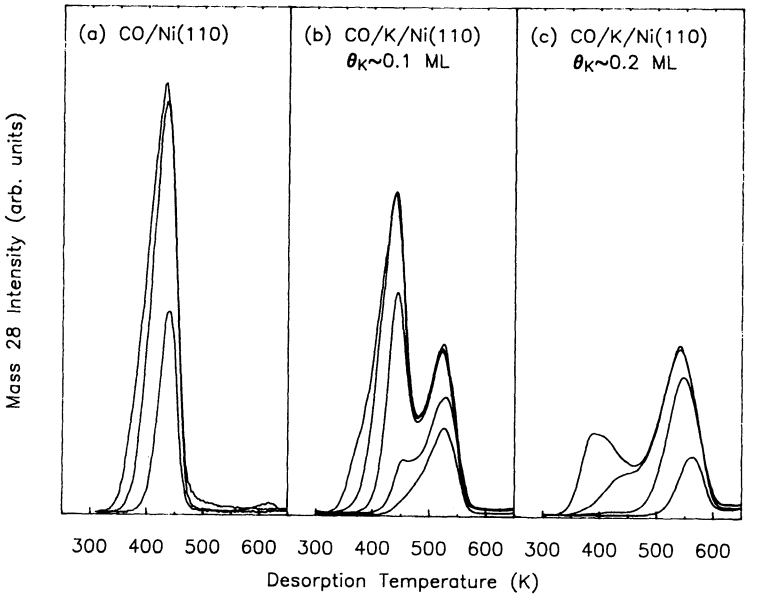


FIGURE 12: Thermal desorption spectra of CO from clean (a) and K-covered (b,c) Ni(110) surface. The adsorption temperature was 300 K, heating rate 13 K/s [39]

species. For a first order desorption reaction (x=1) and a known ratio of pre-exponential factor to heating rate the relationship in equation (8) can be evaluated quantitatively. This evaluation leads to the plot in Fig. 11 which shows the energy of desorption versus peak temperature for different values of k_{ox}/β. First order desorption is usually observed for a species that exhibits molecular (reversible) adsorption. Even if the pre-exponential factor k_o is not known, a change in T_p for different surface conditions (same β) can be taken as a measure of a change in E_d. A nice example is shown in Fig. 12 for CO desorption from a clean and K-covered Ni(110) surface [39]. Whereas the peak temperature for clean Ni(110) is 440 K, the spectra for K-covered Ni(110) exhibit two peaks, one characteristic of CO on clean Ni, and the other characteristic of K-influenced sites at a higher temperature of 525 K and even 560 K, depending on the K-coverage (θ_K). Provided k_o does not change significantly, this increase in T_p of 85-120 K corresponds to an increase in E_d of approximately 20 - 40 kJ/mole according to Fig. 11. This effect illustrates the increase in the adsorption energy of CO due to coadsorbed K [21].

If decomposition or dissociation occurs during adsorption, the order of desorption is generally 2 or even higher. When the desorption order x is unknown, it is still possible to evaluate the energy of desorption from the peak temperature of the flash desorption trace if the heating rate β is varied. For this purpose we rewrite equation (8) such that [38]

$$\ln \frac{\beta}{T_p^2} = \ln \left[\frac{kxn^{x-1} k_{ox}}{E_d}\right] - \frac{E_d}{kT_p} \qquad (9)$$

If desorption traces for equal initial coverages are now recorded with different heating rates, a plot of log (β/T_p^2) versus $1/T_p$ should yield a straight line whose slope can be evaluated to give the energy of desorption. This kind of evaluation has been used in a number of cases where adsorbed molecules decompose during desorption in a rather complicated fashion. A further advantage of the technique related to equation (9) comes from the fact that the pre-exponential factor k_{ox} does not have to be known.

Another important quantity that can be obtained from a flash desorption experiment is the relative or even absolute coverage of the adsorbate. Since the partial pressure of the desorbing species is directly proportional to the rate of desorption, the coverage can be obtained by integrating over the desorption trace:

$$n = \frac{S}{AkT} \int_0^\infty P(t)\, dt \qquad (10)$$

In this equation, S is the pumping speed of the system, A the area of the sample and P(t) is the time-dependent partial pressure recorded during the experiment. Based on a modification of this equation (10), other more complicated schemes for integrating partial desorption traces have been worked out in order to obtain the coverage as a function of temperature for a given initial total coverage. From a complete set of these data, the coverage-dependent desorption energy and pre-exponential factor can, in principle, be calculated [37,40].

Most of the evaluation discussed here works well for reversible first order desorption. Equation (8) has also been applied to second order desorption as long as a single desorption peak is present. The situation be-

comes rather complex when multiple desorption states are present. Similar difficulties arise when desorption is coupled to decomposition of the desorbing species. In those cases, several different desorbing products have to be monitored simultaneously. The question is then whether desorption or decomposition is rate-determining. Several authors have dealt in great detail with questions of this kind [41,42].

4.2. Pre-exponential factor k_o

In order to evaluate the energy of desorption from the peak temperature of a flash desorption trace according to equation (8), a pre-exponential factor of 10^{13} s^{-1} was usually assumed. Independent measurements of the pre-exponential factor of desorbing molecules, such as CO and NO, showed, however, that it was considerably larger than $10^{13}s^{-1}$ and even coverage-dependent. Several authors reported pre-exponential factors exceeding $10^{16}s^{-1}$. A very detailed investigation of CO desorption from a Ru(001) surface by the technique of the heating rate variation resulted in pre-exponential factors between 10^{16} and 10^{19} s^{-1}, as shown in Fig.13 [43].

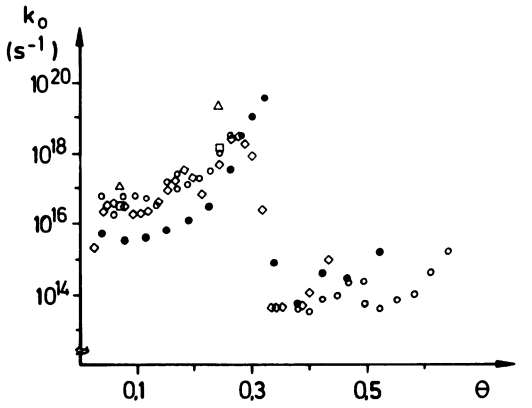


FIGURE 13: Plot of pre-exponential factor k_o versus coverage for CO adsorbed on Ru(001). The data points resulted from different kinds of measurements, such as isothermal desorption and flash desorption [43].

Another similar investigation for CO from a Ni(111) surface reports pre-exponential factors between 10^{17} s^{-1} at very low coverage up to 10^{21} s^{-1} at high coverage [44] . Theoretically these high pre-exponential factors were explained as being due to the internal degrees of freedom of the desorbing molecules. The problem was treated by transition state theory where the pre-exponential factor was then proportional to the ratio of partition functions of the transition state and the adsorbed state. Translational, rotational and vibrational degrees of freedom of the CO molecule were taken into account yielding a pre-exponential factor of about $10^{18}s^{-1}$ at very low coverage [43]. The drawback of this simple theory is that it is not suitable for explaining the coverage dependence of the pre-exponential factor. A somewhat different approach starts from the equilibrium between gas phase and the adsorbed phase resulting in a modified Langmuir isotherm (Fowler equation) [44]

$$P = \frac{\alpha\Theta}{1-\alpha\Theta} \frac{kT}{h^3} (2\pi mkT)^{3/2} \exp\left(-\frac{E_d}{kT}\right) \qquad (11)$$

$\Theta = n/n_0$ is the relative coverage of the adsorbate, h = Planck's constant. Under equilibrium conditions the rate of desorption is equal to the rate of adsorption

$$r_d = r_a = \frac{P}{(2\pi mkT)^{1/2}} \; s(n,T) \qquad (12)$$

and by inserting equation (6) and (11) into equation (12), one obtains an expression for the pre-exponential factor of desorption. One can see that this factor is coverage- and temperature-dependent. The same treatment can be extended to the desorption of molecules by including rotational and vibrational partition functions. Under those assumptions the pre-exponential factor is then given by the following equation:

$$k_o = \frac{\alpha}{n_o(1-\alpha\Theta)} \frac{kT}{h} \; \frac{2\pi mkT}{h^2} \frac{2I_m kT}{h^2} \; s(\Theta,T) \qquad (13)$$

with I_m = moment of inertia. Equation (13) is used semi-empirically in the sense that the experimental data for $s(\Theta,T)$ are inserted. For a sticking coefficient s = 1 and small coverages the pre-exponential factor for CO can be calculated to be of the order of 10^{17} s^{-1} at 300 K, in good agreement with the experiment [44]. Recently another calculation of the pre-exponential factor for NH_3 on Ni(111) surface was carried out in the framework of transition state theory [45].

4.3. Angular and energy dependence

Thus far we have talked about integrated quantities of rates of adsorption and desorption of molecules and therefore ignored any possible angular dependencies of fluxes. Equally we have ignored the energy distribution of desorbing molecules and the question of the internal energy distributions of these molecules. In fact, asking questions about the angular and energy dependence of desorbing molecules, is the point where the real science of desorption begins. Under the assumption of thermodynamic equilibrium, the angular distribution of the flux, $F(\Theta,\varphi)$ of molecules leaving from a surface area element dA is given by the following expression (Knudsen) [46]:

$$F(\vartheta,\varphi)\; dA = \frac{n_v}{4\pi} \left(\frac{8kT}{\pi m}\right)^{1/2} \cos\vartheta \; dA \qquad (14)$$

In equation (14) n_v is the concentration of molecules per unit volume (cm^{-3}), ϑ the polar angle and φ the azimuth. Whereas the probability of desorption does not depend on the azimuth, it does depend on the polar angle ϑ and is proportional to cos ϑ. This means that there is a maximum in the rate of desorption normal to surface.

The velocity (energy) dependence of molecules (flux) leaving the surface in equilibrium can also be given by a simple expression. The flux of molecules d^3f normalized to unit solid angle desorbing with velocities between v and v + dv are given by [46]:

$$\frac{d^3f}{\sin\vartheta \; d\varphi \; d\vartheta} = n_v \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{mv^2}{2kT}\right) v^3 \cos\vartheta \; dv \qquad (15)$$

This equation describes the Maxwell distribution. Of some importance is the root mean square of the velocity given by [46]

$$v_{rms} = \left(\frac{4kT}{m}\right)^{1/2} \tag{16}$$

This velocity is related to the mean kinetic energy of desorbing molecules:

$$\langle E \rangle = \frac{1}{2} m v_{rms}^2 = 2kT \tag{17}$$

The laws for angular and velocity distributions, equations (14) and (15), have been derived, under the assumption of equilibrium and microscopic reversibility, for all molecules leaving the surface. Here it is important to realize that the flux of molecules coming from the surface at equilibrium includes those species that were actually adsorbed and those that are scattered or reflected. When the sticking coefficient $s(T,\vartheta,n)$ is equal to unity, there are no reflected but only adsorbing molecules. In that case the desorption flux consists of desorbing molecules only, and eqs. (14) and (15) are a correct description for the angular and velocity distributions of the desorbing species. In the general case of $s(t,\vartheta,n) < 1$, however, there is at present no analytical expression for the description of the angle and velocity dependence of molecules desorbing from a surface. In this more complicated case one can derive a relationship between the desorption and adsorption probabilities which can be tested experimentally. The physical law of microscopic reversibility requires that desorption and adsorption should show the same angular dependence. Assuming thermodynamic equilibrium, one obtains the relationship [12]:

$$D(T,\vartheta,n) = s(T,\vartheta,n) \cos\vartheta \tag{18}$$

where $D(T,\vartheta,n)$ is the desorption probability. A test of this relationship under the conditions of thermodynamic equilibrium (adsorption and desorption taking place simultaneously) is practically impossible. Hence, as a compromise, experiments are performed under conditions of quasi-equilibrium, checking the probabilities of adsorption and desorption in separate experiments [12]. This procedure will be illustrated by examples.

There is a large difference in sticking probability for the adsorption of CO and H_2 on a Ni(111) surface [10]. Whereas CO adsorbs with an initial sticking coefficient near unity, that of H_2 is only 0.05. Therefore, the adsorption/desorption of these gases on Ni(111) was chosen to check eq. (18). The result for desorption of CO from a Ni(111) surface is shown in Fig. 14 [12]. Here, the desorption probability as well as the adsorption probability is plotted versus polar angle ϑ. One can see that the experimental data obey equation (14) very well and exhibit a cosine distribution. For comparison, Fig. 15 shows the adsorption and desorption probabilities versus polar angle for H_2 on Ni(111). In this case, the distribution deviates very substantially from a simple cosine law. It is fitted with a $\cos^x\vartheta$ function where $x = 4.5$. There is still agreement in the angular dependence of the normalized adsorption and desorption probabilities suggesting that the condition of quasi-equilibrium is fulfilled. For the same data it was also demonstrated that the desorption of H_2 did not obey the Maxwell law. The mean energy of desorbing H_2 molecules was considerably higher than the corresponding surface temperature of the Ni crystal [47].

The point in a comparison of CO and H_2 adsorption/desorption from a Ni(111) surface is that CO adsorbs reversibly and without any hint of an activation barrier for adsorption, while H_2 adsorbs dissociatively with considerable evidence for an activation barrier of adsorption. The angular velocity distribution of H_2 from a Ni(111) surface has also been studied in detail by time-of-flight mass spectroscopy [47]. These experiments were carried out for clean- and sulfur-covered surfaces. Typical energy

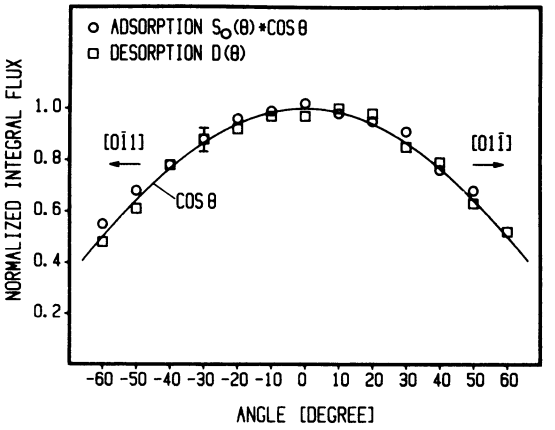


FIGURE 14: Angle dependence of the adsorption and desorption probability of CO on Ni(111) at 300 K [12].

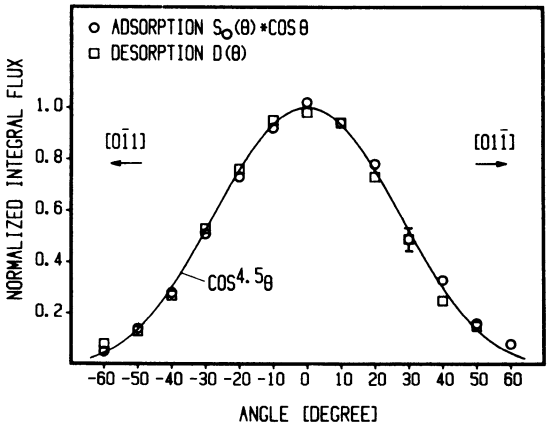


FIGURE 15: Angle dependence of the adsorption and desorption probability of H_2 on Ni(111) at 190 K [12]. The adsorbing gas has a temperature of 300 K (Knudsen cell).

distributions are shown in Fig. 16 for three different polar angles ϑ. In each case the temperature of the Ni crystal was 1143 K. The mean energy of each distribution is given as $\langle E\rangle/2k$ which deviates considerably from the surface temperature. For comparison, Maxwell distributions with the surface temperature and a temperature corresponding to $\langle E\rangle/2k$ are shown in each one of the figures. It is clear that the experimental distributions are non-Maxwellian. It turns out that molecules desorbing normal to the surface ($\vartheta = 0$) have the highest mean temperature, i.e. 1871 K, whereas those desorbing at larger polar angles have lower temperatures, e.g. at 80° only 771 K. The angular dependence of the mean energy of desorbing D_2 molecules from Ni(111) at a surface temperature of 1134 K is summarized in Fig. 17. The reason for the non-applicability of eq. (15) comes from the

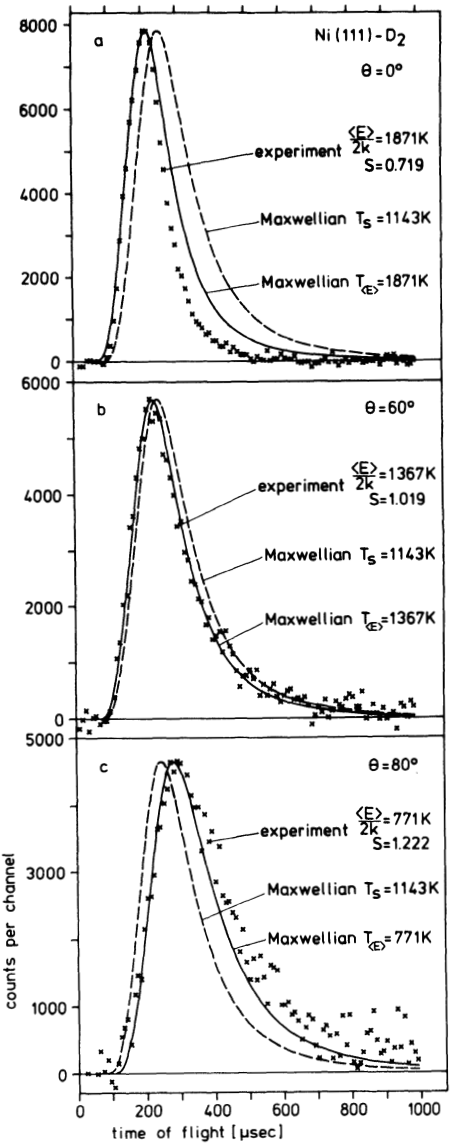


FIGURE 16: Time-of-flight spectra of D_2 molecules desorbing from a S-covered Ni(111) surface at polar angles of 0°, 60° and 80°. The curves represent Maxwell spectra corresponding to T = 1143 K (surface temperature) and $T_{<E>} = <E>/2k$ [47].

fact that there is an activation barrier for adsorption in this case. Even though the barrier is not known in any detail, one may infer that molecules desorbing have crossed this barrier and hence are more energetic than corresponding to the surface temperature. Since the barrier is a true surface property, molecules having crossed it no longer have an opportunity to lose energy to the solid. Hence they carry along the extra energy necessary to surmount the barrier as translational energy.

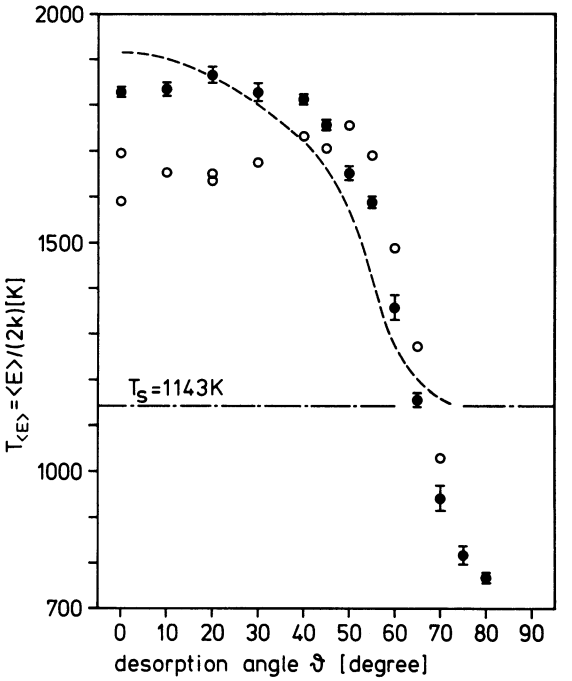


FIGURE 17: Polar angle dependence of mean energy of D_2 molecules desorbing from clean (o) and S-covered (•) Ni(111) [47].

Another interesting example in this connection concerns the desorption of D_2 from a Pd(100) surface [48]. Fig. 18 shows again a set of energy distribution curves of desorbing D_2 at a surface temperature of 500 K and for $\vartheta = 0^0$. The distributions shown represent the clean surface as well as surfaces with an increasing amount of adsorbed sulfur. The peculiar feature is now that the distribution for the clean Pd surface is Maxwellian whereas the distributions with a larger amount of sulfur on the surface are strongly non-Maxwellian. For example, the distribution for a sulfur coverage of 0.65 is characterized by a very narrow peak at high energy corresponding to a temperature of about 3 times the surface temperature. Hence one could conclude that D_2 desorption from clean Pd is not influenced by an activation barrier of adsorption, but that on sulfur-covered Pd it is. As a rationale for these experimental data, a one-dimensional potential energy diagram shown in Fig. 19 was proposed [46]. Since hydrogen dissolves easily in Pd, the energy diagram continues into the bulk of the crystal. On clean Pd the chemisorption state of hydrogen is characterized by an energy of about 20 kcal/mol. Hydrogen penetrating the bulk of a crystal and desorbing from the surface equilibrates in the chemisorption well, then recombines to form the H_2 molecule and desorbs. In the absence of an activation barrier of adsorption this desorption obeys the Knudsen and Maxwell distributions in equations (14) and (15). On the other hand, when sulfur is adsorbed on the Pd surface, the adsorption of hydrogen in its normal chemisorption state is poisoned, i.e. the chemisorption well in Fig. 19 is no longer accessible. Therefore, hydrogen coming from the bulk of the Pd, will

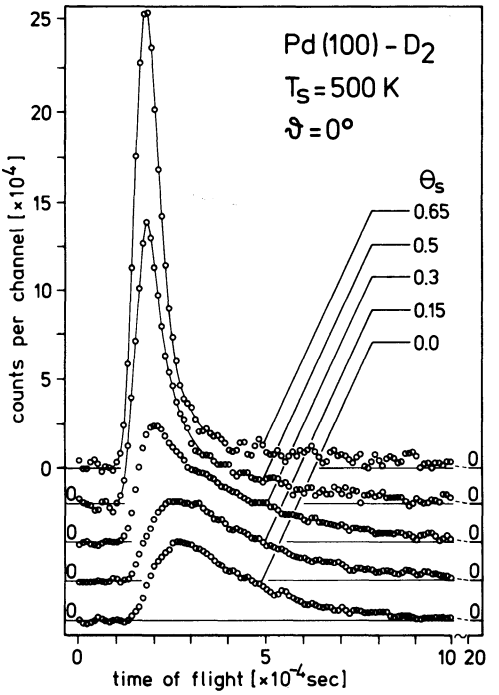


FIGURE 18: Time-of-flight spectra of D_2 molecule desorbing from clean and S-covered Pd(100) at 500 K (surface temperature). The sulfur coverage is given by Θ_s [48].

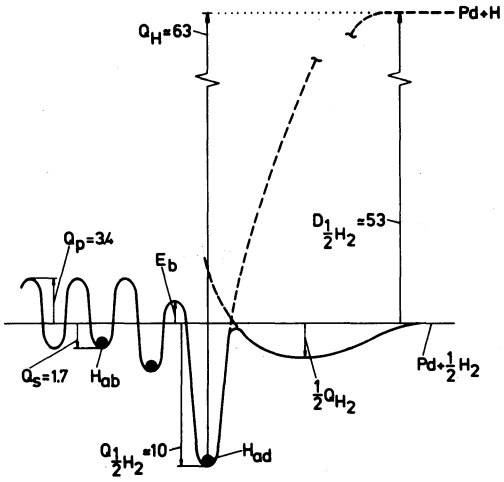


FIGURE 19: Energy diagram for H_2 on Pd, showing a molecular (H_2) and atomic H_{ad} adsorption state as well as bulk absorption states (H_{ab}). E_b last bulk barrier; Q_p permeation energy; Q_s solution energy (all energies in (kcal/mol) [46].

equilibrate in the last well of the bulk potential and then recombine to form H_2 molecules near the top of the last barrier and then desorb. Because of the existence of this bulk potential barrier, E_b, the desorbing H_2 molecules will have again a high translational energy giving rise to the narrow peak in the energy distribution of Fig. 18. This example then underlines again the correlation of activated adsorption and the deviations from Maxwell and Knudsen distributions.

4.4. Internal energy of desorbing molecules

The naive point of view that the internal energy of desorbing molecules should correspond to the surface temperature of the substrate is generally incorrect. The reasoning is quite analogous to that in the previous section. Again it is useful to distinguish between non-activated adsorption on the one hand, and activated dissociative adsorption on the other hand. In the first case, the internal energies may correspond to the surface temperature, but in the second case considerable deviations can be anticipated. The principal reason for these deviations is the activation barrier of adsorption. The recombination of atoms into a molecule occurs near the top of the barrier, and any excess (or deficiency) in internal energy associated with this event will be carried over into the gas phase because the desorbing molecule has ceased to interact with the surface. Hence, the molecule might desorb in an excited vibrational or rotational state. The same occurs when desorbing molecules are formed as a product of a surface reaction, e.g. in the case of CO oxidation.

An interesting example is the desorption of N_2 from a polycrystalline Fe surface [49]. Here the vibrational state of desorbing N_2 molecules was measured by electron beam induced fluorescence. The vibrational energy of the molecule was then converted into an equivalent vibrational temperature. The experiments were carried out for surface temperatures between 1080 and 1400K. The vibrational temperature is shown as a function of the surface temperature in Fig. 20 for a clean Fe surface. Here the difference between

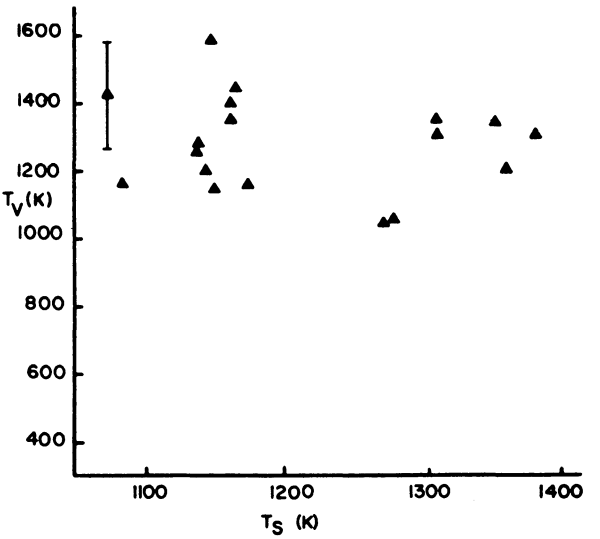


FIGURE 20: Vibrational temperature versus surface temperature for N_2 desorbing from clean Fe [49].

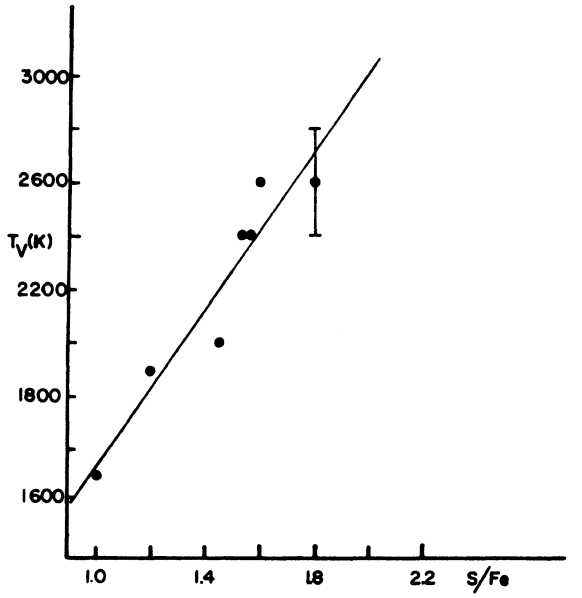


FIGURE 21: Vibrational temperature of desorbing N_2 molecules versus S/Fe Auger peak ratio for a surface temperature of Fe of 1150 K [49].

these two temperatures does not seem to be very significant. On the other hand, sulfur-contaminated Fe surfaces gave rise to very large vibrational temperatures of desorbing N_2 molecules, as seen in Fig. 21. In this figure, the vibrational temperature is plotted versus the S/Fe Auger peak ratio, i.e. versus sulfur surface coverage. Note the strong increase in vibrational temperature with increasing sulfur coverage. These data are rationalized in terms of an increased activation barrier for N_2 adsorption on a sulfur-covered Fe surface [49].

The rotational state of desorbing N_2 molecules was also studied by the same group [49]. It was found that the rotational temperature was about 400 ± 50 K independent of the surface temperature of the Fe or of its state of cleanliness. The low value of the rotational temperature seems to indicate that the recombination of N atoms into a N_2 molecule is sterically hindered. This steric hinderence must be a consequence of the activation barrier where the recombination of N atoms to the N_2 molecule occurs.

5. Surface reactions

One of the interesting aspects of studying adsorption and desorption of gases from solid surfaces is the possibility that a heterogeneous surface reaction might take place. This aspect has a considerable technological relevance. We can distinguish various types of surface reactions:

a) The first kind is dissociative adsorption which we have already learned about by a few examples.
b) The second type is dissociative desorption or decomposition. This means that on raising the temperature, the adsorbed species begin to decompose

just prior to desorption, sometimes concurrently with desorption. For example, the desorption and decomposition of formic acid which has been studied in great detail may occur as follows [50]:

$$HCOOH \begin{cases} \nearrow CO_2 + H_2 \\ \searrow CO + H_2O \end{cases} \qquad (19)$$

It is a challenge for the surface scientist to find out when and why the decomposition occurs via the first or the second channel. Many other more complicated examples of this type have also been investigated [51].

c) A more significant example is a bimolecular heterogeneous reaction such as the oxidation of CO to CO_2:

$$CO_{ad} + O_{ad} \rightarrow CO_{2ad} \rightarrow CO_2 \qquad (20)$$

This reaction occurs by a so-called Langmuir-Hinshelwood mechanism in which both reactants are coadsorbed on the surface [52]. In this particular example the reactants are CO and O which requires the dissociative adsorption of oxygen as a primary step. Many different investigations of this reaction have appeared in the literature [52].

d) In conjunction with this bimolecular heterogeneous reaction attempts have been undertaken to search for a mutual bond weakening/strengthening of the coadsorbed reactants, an effect, which might be interpretable as a precursor situation to an actual surface reaction. This search involved the investigation of stationary adsorbed phases by electron spectroscopies, for example, but proved to be unsuccessful in most cases.

The general experimental approach to studying surface reactions is similar to that for studying adsorption and desorption of gases. Experiments are usually carried out on well characterized clean single crystal or polycrystalline surfaces by utilizing surface spectroscopic techniques and working under ultra high vacuum conditions. Of course, the study of the steady-state surface reaction requires the presence of reasonably high partial pressures of the reactants inside the vacuum system. Under such conditions it is sometimes difficult to maintain and guarantee a high purity of the gas mixture. This problem can be overcome by using molecular beams of the reactant molecules. In some cases the reaction probability is extremely low, however, such that it can no longer be studied inside a vacuum system. A rather well-known example is the hydrogenation of CO on transition metal surfaces which occurs with a measurable rate at pressures of at least several mbar. In order to study such a reaction, a combination system is generally used featuring a high-pressure microreactor, a fast transfer system between reactor and vacuum, and a ultrahigh vacuum system for surface spectroscopy. The experimental results consist of reaction-kinetic data obtained by mass spectrometry or by a gas chromatograph and by surface analytical data.

The hydrogenation of CO on transition metal surfaces is an important example of a surface reaction because of its technological relevance [53]. It is also a very complicated reaction in terms of mechanism. The spectrum of products obtained from this reaction is extremely broad, ranging from simple methane to long-chain hydrocarbons, alcohols and other oxygen-containing molecules. The selectivity of this reaction depends on a large number of variables, such as temperature, reactant composition, gas flow, and surface composition. Surface additives such as alkali metals have a tremendous influence on the selectivity and reactivity of the catalyst [53]. Figure 22 shows as an example C1s and K2p spectra of a polycrystal

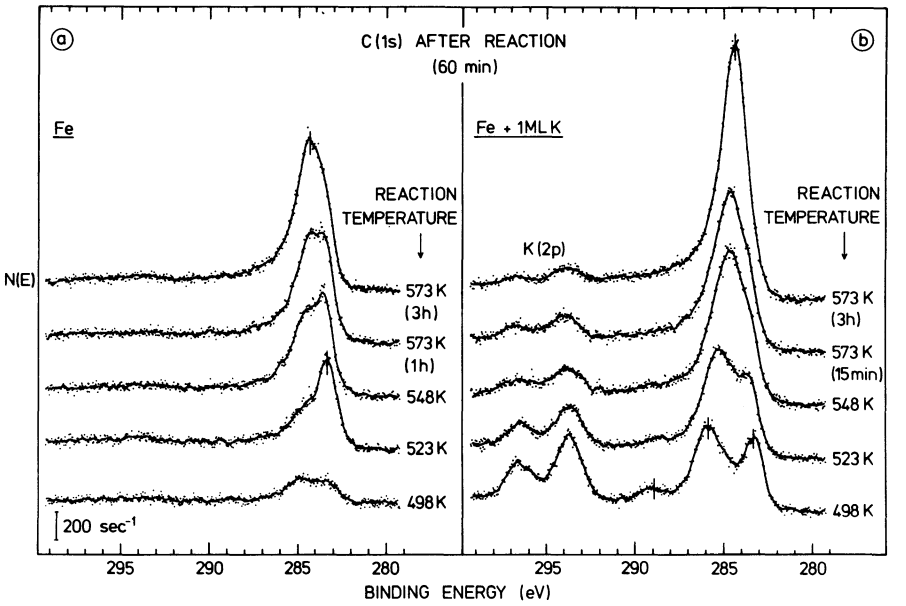


FIGURE 22: C1s and K2p XPS spectra after CO hydrogenation on Fe at various temperatures as indicated. The reaction time was 60 min except where otherwise indicated. Past (a) was initially clean Fe, part (b) Fe with one monolayer of K [54].

line Fe foil after CO hydrogenation at various temperatures. The reaction took place in a flow system at 1 bar total pressure and a partial pressure ratio CO/H_2 of 1:20, the reaction time in each case was 60 min [54]. The left hand side of the figure shows C1s spectra for the unpromoted (K-free) surface while the right hand side is characteristic of a surface with about one monolayer of pre-adsorbed K evaporated onto the clean Fe inside the UHV chamber. The C1s spectra as a function of reaction temperature for clean and K-covered Fe demonstrate the dependence of the surface composition of this catalyst on running conditions. In some cases there are several distinguishable C1s contributions which may be correlated with different carbon-containing adsorbed species. The identification of the various C1s peak contributions in Fig. 22 was accomplished by a comparison of binding energies with layers of known composition, such as simple carbidic (atomic) and graphitic carbon, adsorbed CO, or adsorbed hydrocarbon molecules [55]. The shape of the differentiated carbon Auger peak was also taken as an additional aid in identifying the carbonaceous species. With that knowledge the C1s peaks at 283.3 eV and 284.5 eV binding energy in Fig. 22 were attributed to carbidic and graphitic carbon, respectively. The additional peaks at 285.8 eV and 289.0 eV observed for the K-covered Fe were interpreted as being due to a polymerized surface carbon species ("Polymethylene") and a carbonate-type species, respectively. No detailed knowledge of these latter species (structure, stoichiometry, etc.) exists. Additional more sensitive analytical techniques are required in order to analyze these species in more detail and also to detect minority species which must be present as reaction intermediates. The remarkable feature in

the results of Fig. 22 is that the surface carbon composition is indeed different for the clean and K-promoted Fe, and that the K causes the rate of carbon deposition to increase [56,57]. That could be shown by integrating the total C1s intensity as a function of reaction time.

ACKNOWLEDGEMENT

I am very grateful to Prof. George Comsa for clarifying discussions and critical comments on the manuscript.

REFERENCES

1. D.O. Hayward and B.M.W. Trapnell: Chemisorption (Butterworths, London 1964).
2. G. Ehrlich: J. Phys. Chem. Solids 1 (1956) 1.
3. J.E. Lennard-Jones: Trans.Faraday Soc. 28 (1932) 333.
4. D.A. King and M.G. Wells: Surface Sci. 29 (1972) 454.
5. H. Steininger, S. Lehwald and H. Ibach: Surface Sci. 123 (1982) 264.
6. G. Ertl, M. Neumann and K.M. Streit: Surface Sci. 64 (1977) 393.
7. C.T. Campbell, G. Ertl, H. Kuipers and J. Segner: Surface Sci. 107 (1981) 207.
8. P. Kisliuk: J. Phys. Chem. Solids 3 (1957) 95; and J. Phys. Chem. Solids 5 (1958) 78.
9. K. Schönhammer: Surface Sci. 83 (1979) L 633.
10. A. Winkler and K.D. Rendulic: Surface Sci. 118 (1982) 19
11. H.J. Robota, W. Vielhaber, M.C. Lin, J. Segner and G. Ertl: Surface Sci. 155 (1985) 101.
12. H.P. Steinrück, K.D. Rendulic and A. Winkler: Surface Sci. 154 (1985) 99.
13. S.P. Singh-Boparai, M. Bowker and D.A. King: Surface Sci. 53 (1975) 55.
14. P.W. Tamm and L.D. Schmidt: J. Chem. Phys. 54 (1971) 4775.
15. G. Pirug, G. Brodén and H.P. Bonzel: Proc. 7th Intl. Vac. Congr. and 3. Intl. Conf. on Solid Surf., p. 907, Vienna 1977.
16. H.P. Bonzel, G. Brodén, and G. Pirug: J. Catalysis 53 (1978) 96.
17. J. Perdereau and G. E. Rhead: Surface Sci. 24 (1971) 555; B. Lang, R.W. Joyner and G.A. Somorjai: Surface Sci. 30 (1972) 454.
18. H. Ibach, K. Horn, R. Dorn and H. Lüth: Surface Sci. 38 (1973) 433.
19. K. Christmann and G. Ertl: Surface Sci. 60 (1976) 365.
20. K. Besocke and K. Berger: Proc. 7. Intl. Vac. Congr. and 3. Intl. Conf. on Solid Surf., p. 893, Vienna 1977.
21. H.P. Bonzel: J. Vac. Sci. Technol. A2 (1984) 866.
22. J.C. Tracy and P.W. Palmberg: Surface Sci. 14 (1969) 274.
23. J.A. Van den Berg, L.K. Verheij and D.G. Armour, Surface Sci. 91 (1980) 218; H. Niehus and G. Comsa, Surface Sci. 151 (1985) L171.
24. K.H. Rieder and T. Engel, Phys. Rev. Letters 43 (1979) 373; K. Christmann, F. Chehab, V. Penka and G. Ertl, Surface Sci. 152/153 (1985) 356.
25. R.P.N. Bronckers and A.G.J. de Wit, Surface Sci. 112 (1981) 133; H. Niehus and G. Comsa, Surface Sci. 140 (1984) 18.
26. B.E. Hayden, K.C. Prince, P.J. Davie, G. Paolucci and A.M. Bradshaw: Solid State Comm. 48 (1983) 325.
27. H. Ibach and D.L. Mills: Electron Energy Loss Spectroscopy and Surface Vibrations (Academic Press, New York, 1982).
28. J. Stöhr and R. Jaeger: Phys. Rev. B 26 (1982) 4111.
29. K.A. Thompson and C.S. Fadley: J. Electr. Spectr. Rel. Phen. 33 (1984) 29.

30. F. Sette, J. Stöhr; E.B. Kollin, D.J. Dwyer, J.L. Gland, J.L. Robbins and A.L. Johson: Phys. Rev. Letters 54 (1985) 935.
31. D.A. Wesner, F.P. Coenen and H.P. Bonzel: Phys. Rev. B 33 (1986) 8837.
32. G. Brodén, G. Gafner and H.P. Bonzel: Surface Sci. 84 (1979) 295.
33. R. Gomer: Disc. Faraday Soc. 28 (1959) 23.
34. G. Binnig and H. Rohrer: Physica 127 B (1984) 37.
35. B. Poelsema, L.K. Verheij and G. Comsa: Phys. Rev. Letters 49 (1982) 1731.
36. C.P. Slichter: Bull. Am. Phys. Soc. 31 (1986) 534.
37. D.A. King: Surface Sci. 47 (1975) 384.
38. P.A. Redhead: Vacuum 12 (1962) 203.
39. L.J. Whitman and W. Ho: J. Chem. Phys. 83 (1985) 4808.
40. E. Bauer, F. Bonczek, H. Poppa and G. Todd: Surface Sci. 53 (1975) 87.
41. R. Gorte and L.D. Schmidt: Appl. Surf. Sci. 3 (1979) 381.
42. A. Cassuto and D.A. King: Surface Sci. 102 (1981) 388.
43. H. Pfnür, P. Feulner, H.A. Engelhardt and D. Menzel: Chem. Phys. Letters 59 (1978) 481; H. Pfnür and D. Menzel: J. Chem. Phys. 79 (1983) 2400.
44. H. Ibach, W. Erley and H. Wagner: Surface Sci. 92 (1980) 29.
45. A. Redondo, Y. Zeiri, J.J. Low and W.A. Goddard III: J. Chem. Phys. 79 (1983) 6410.
46. G. Comsa and R. David: Surface Sci. Reports 5 (1985) 145.
47. G. Comsa, R. David and B.J. Schumacher: Surface Sci. 85 (1979) 45.
48. G. Comsa, R. David and B.J. Schumacher: Surface Sci. 95 (1980) L210.
49. R.P. Thorman and S.L. Bernasek: J. Chem. Phys. 74 (1981) 6498.
50. J.L. Falcomer and R.J. Madix: Surface Sci. 48 (1975) 393.
51. R.J. Madix: in The Chemical Physics of Solid Surfaces and Heterogeneous Catalysis, Vol. 4, ed. by D.A. King and D.P. Woodruff (Elsevier, Amsterdam, 1982) chapters 1 and 4.
52. T. Engel and G. Ertl: in Ref. 51, chapter 3.
53. R.B. Anderson: in Catalysis, ed. by P.H. Emmett (Reinhold, New York, 1956), Vol. 4.
54. D.A. Wesner, F.P. Coenen and H.P. Bonzel: Langmuir 1 (1985) 478.
55. H.P. Bonzel and H.J. Krebs: Surface Sci. 91 (1980) 499.
56. H.P. Bonzel and H.J. Krebs: Surface Sci. 109 (1981) L527.
57. H.P. Bonzel, G. Brodén and H.J. Krebs: Appl. Surface Sci. 16 (1983) 373.